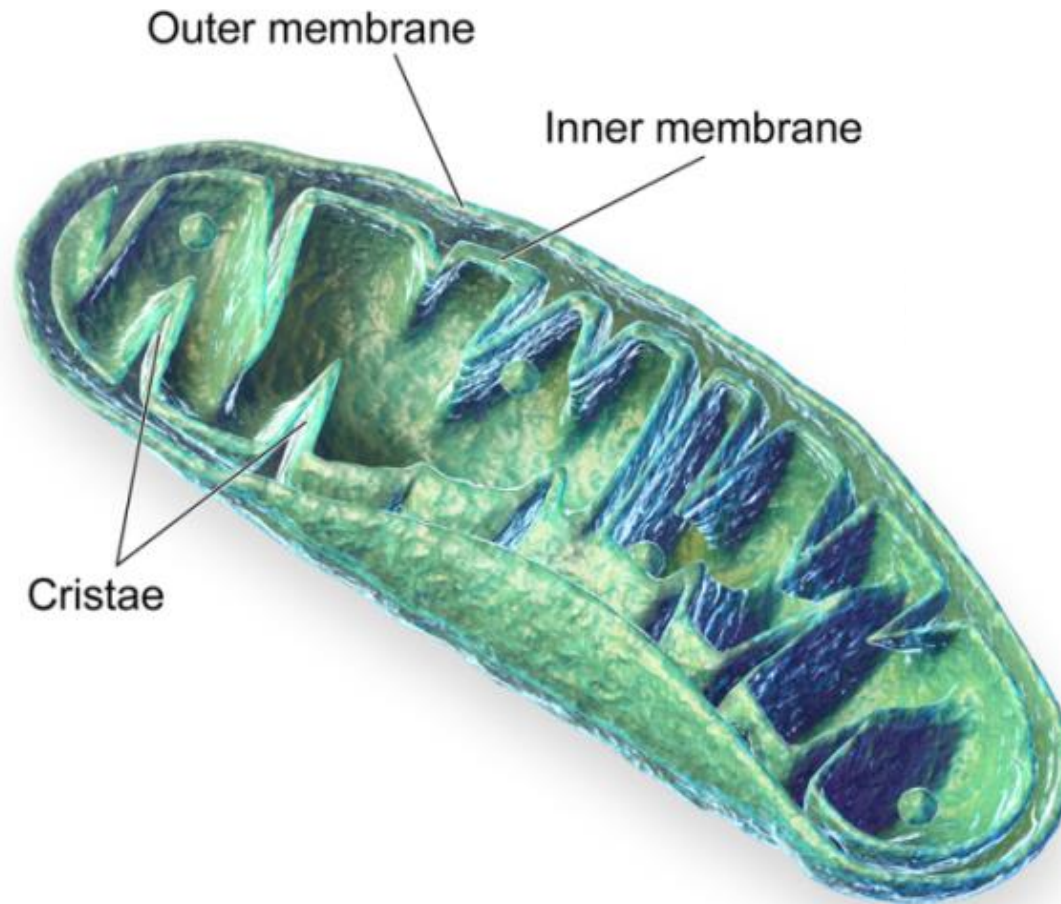


Electron Transport Chain

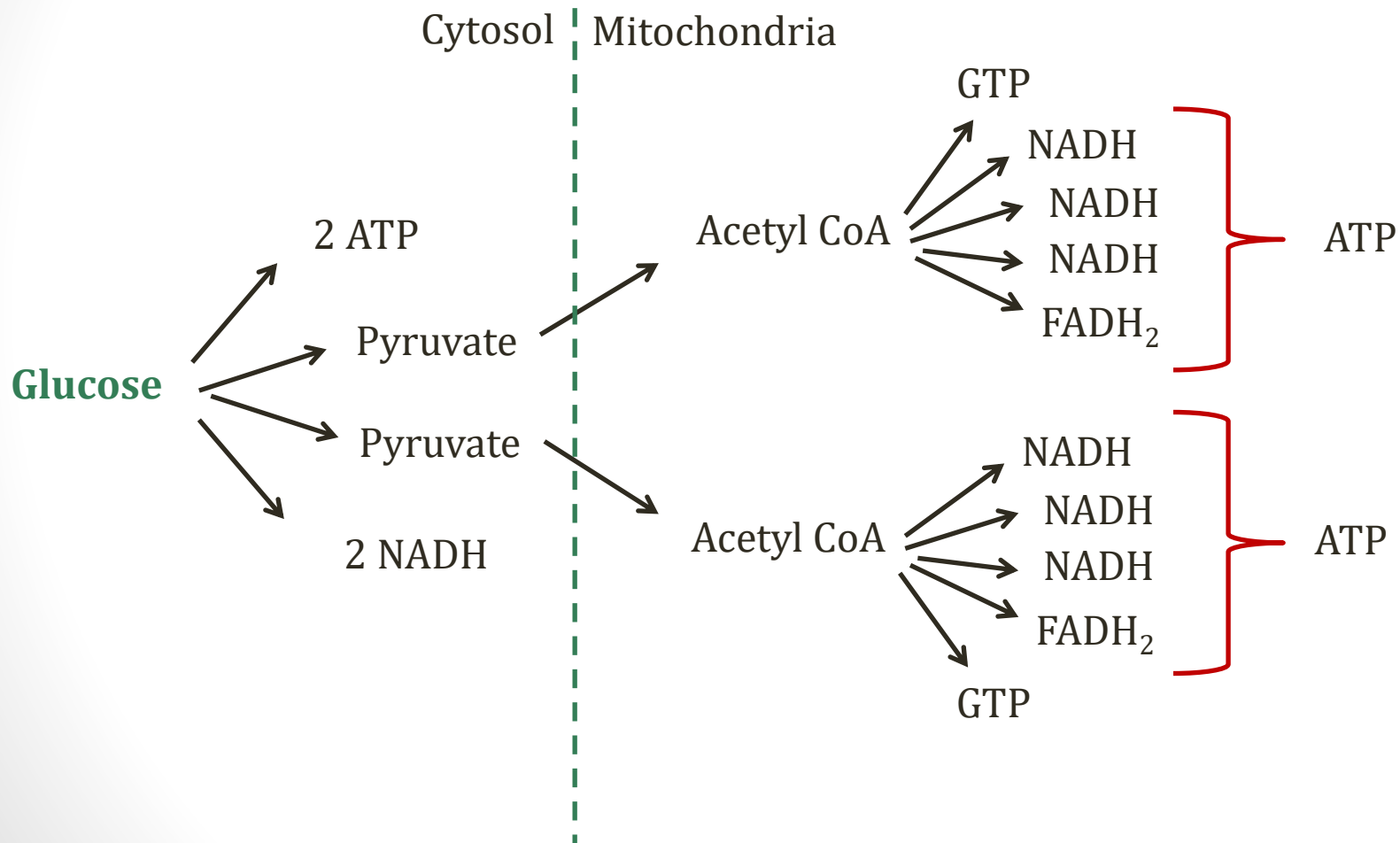
Jason Ryan, MD, MPH

Electron Transport Chain

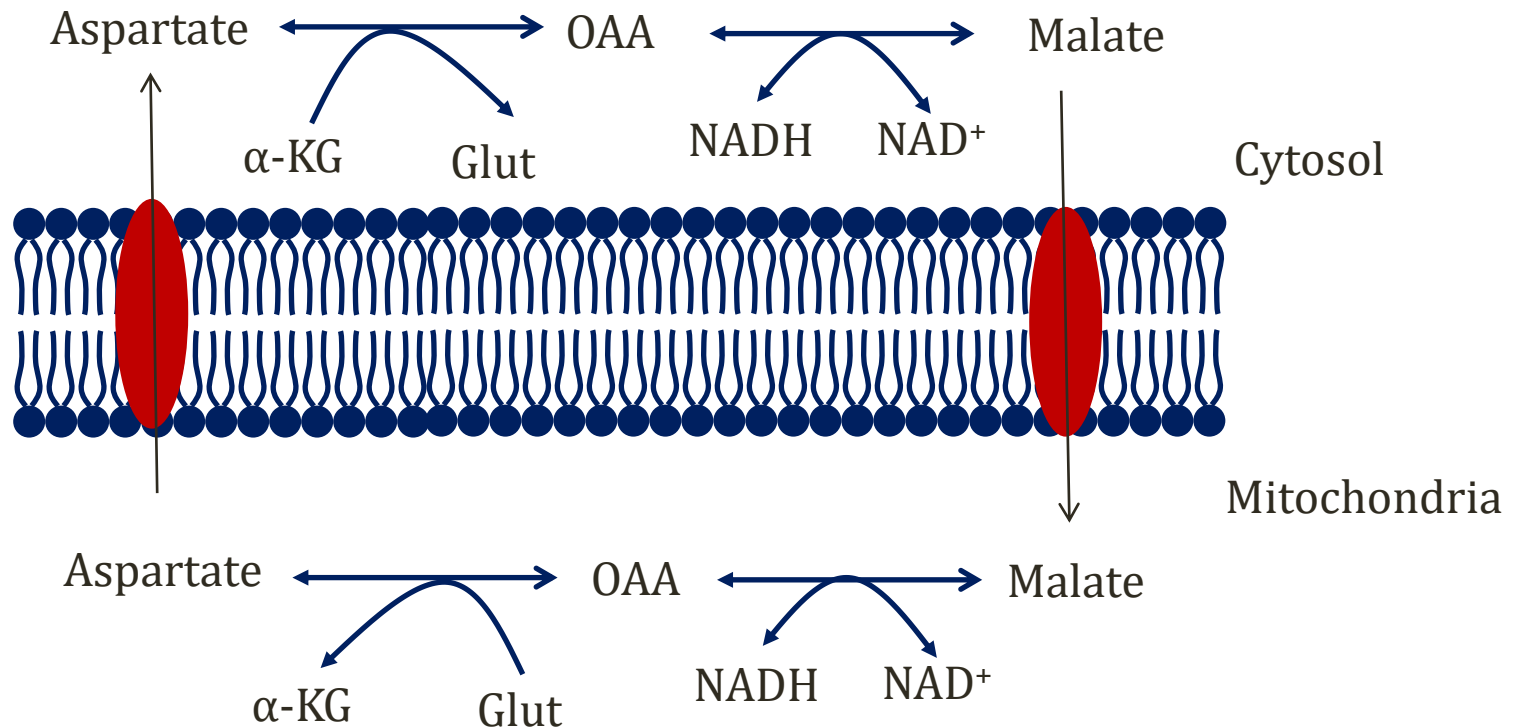


NADH
FADH₂

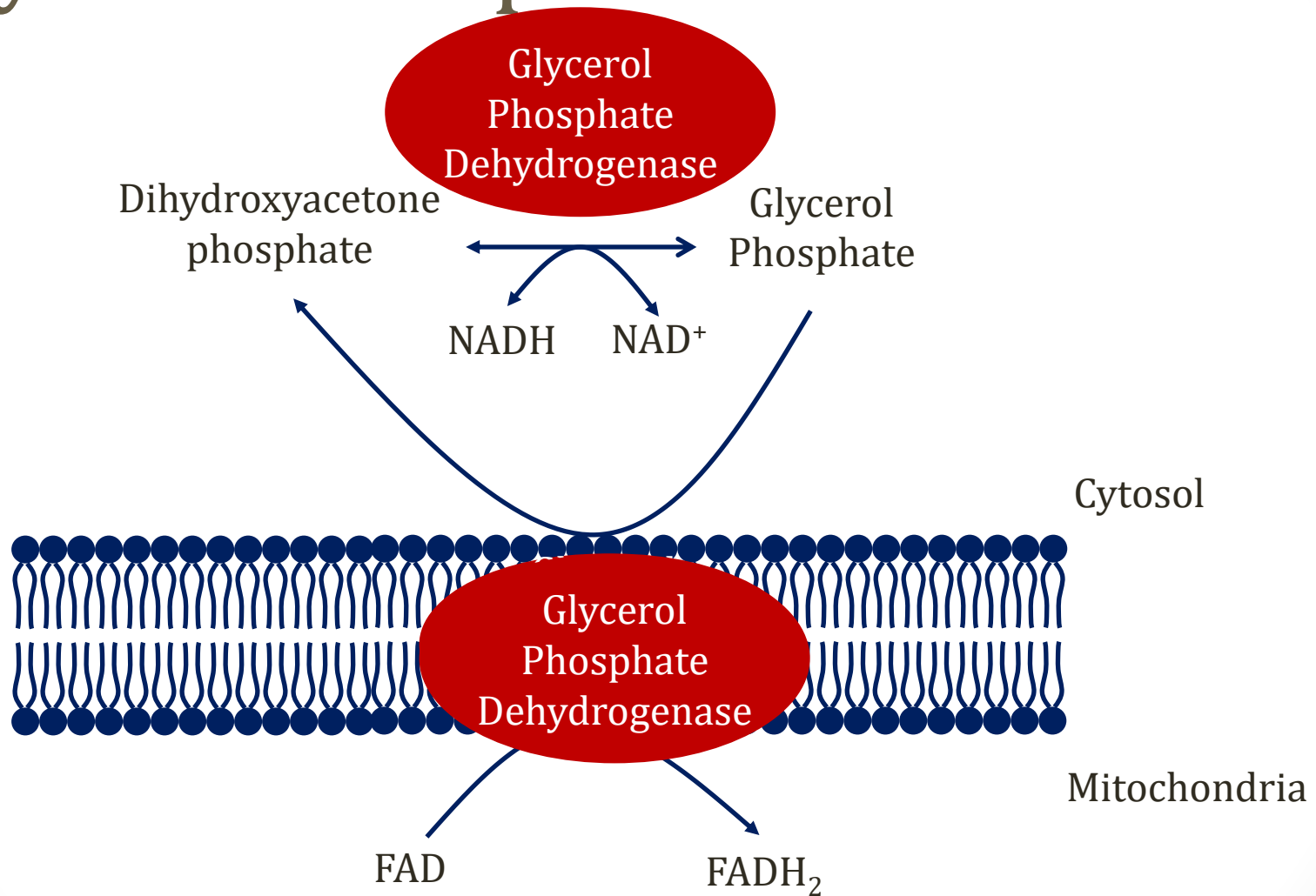
Aerobic Metabolism



Malate Shuttle

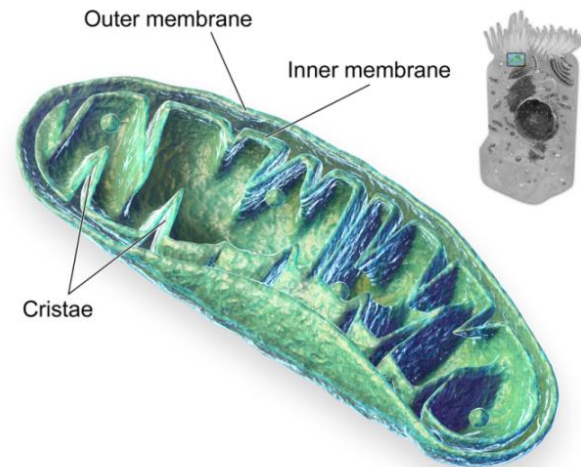


Glycerol Phosphate Shuttle



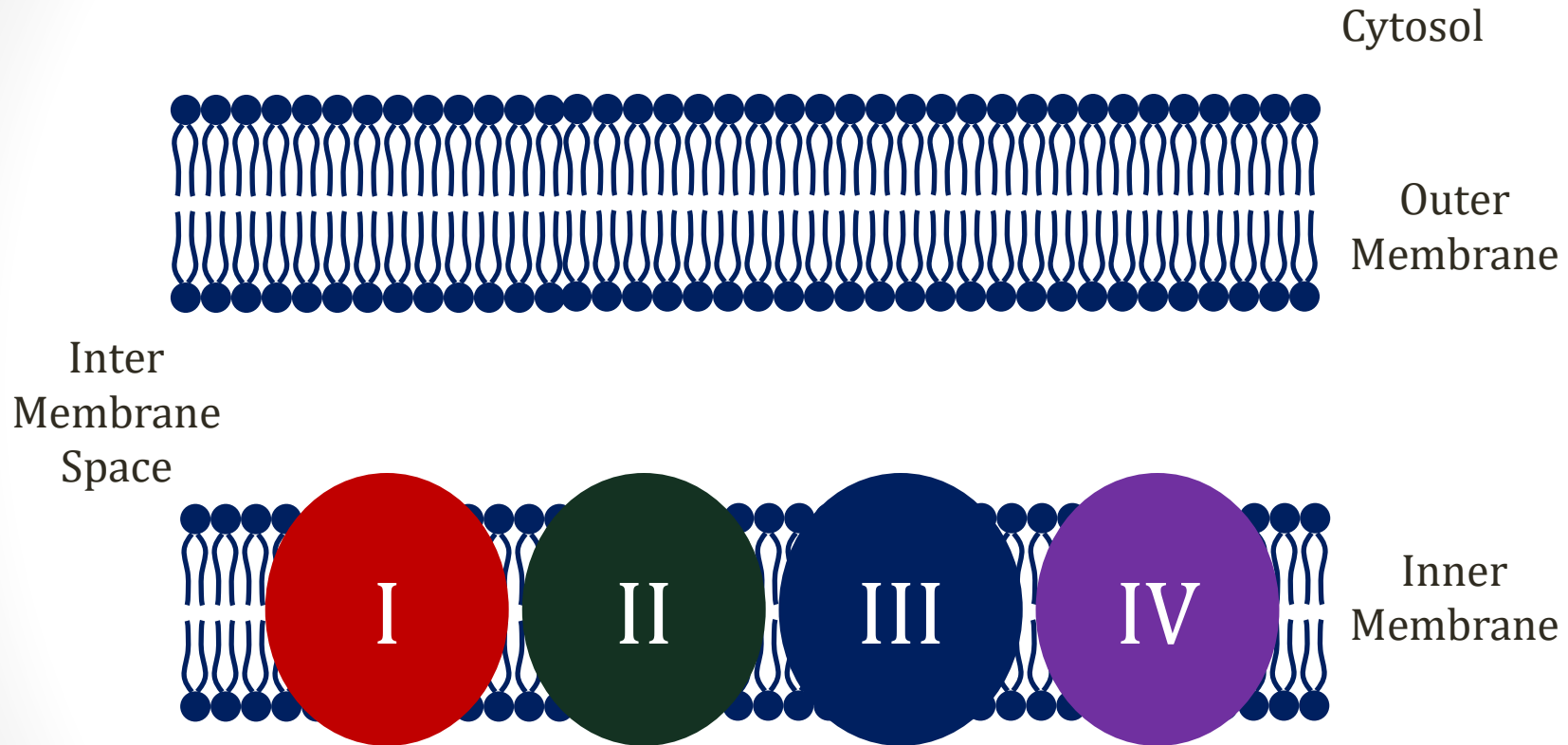
Electron Transport

- Extract electrons from NADH/FADH₂
- Transfer to **oxygen** (*aerobic* respiration)
- In process, generate/capture energy
- $\text{NADH} \rightarrow \text{NAD}^+ + \text{H}^+ + 2\text{e}^-$
- $\text{FADH}_2 \rightarrow \text{FAD} + 2\text{H}^+ + 2\text{e}^-$
- $2\text{e}^- + 2\text{H}^+ + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$



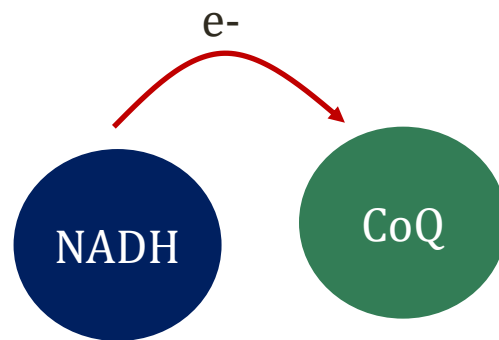
Mitochondria

Electron Transport Complexes



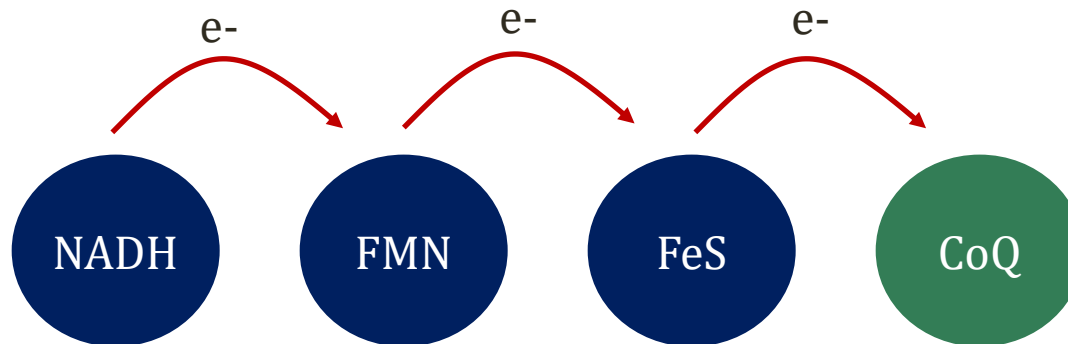
Complex I

- **NADH Dehydrogenase**
- Oxidizes **NADH** ($\text{NADH} \rightarrow \text{NAD}^+$)
- Transfers electrons to **coenzyme Q** (ubiquinone)

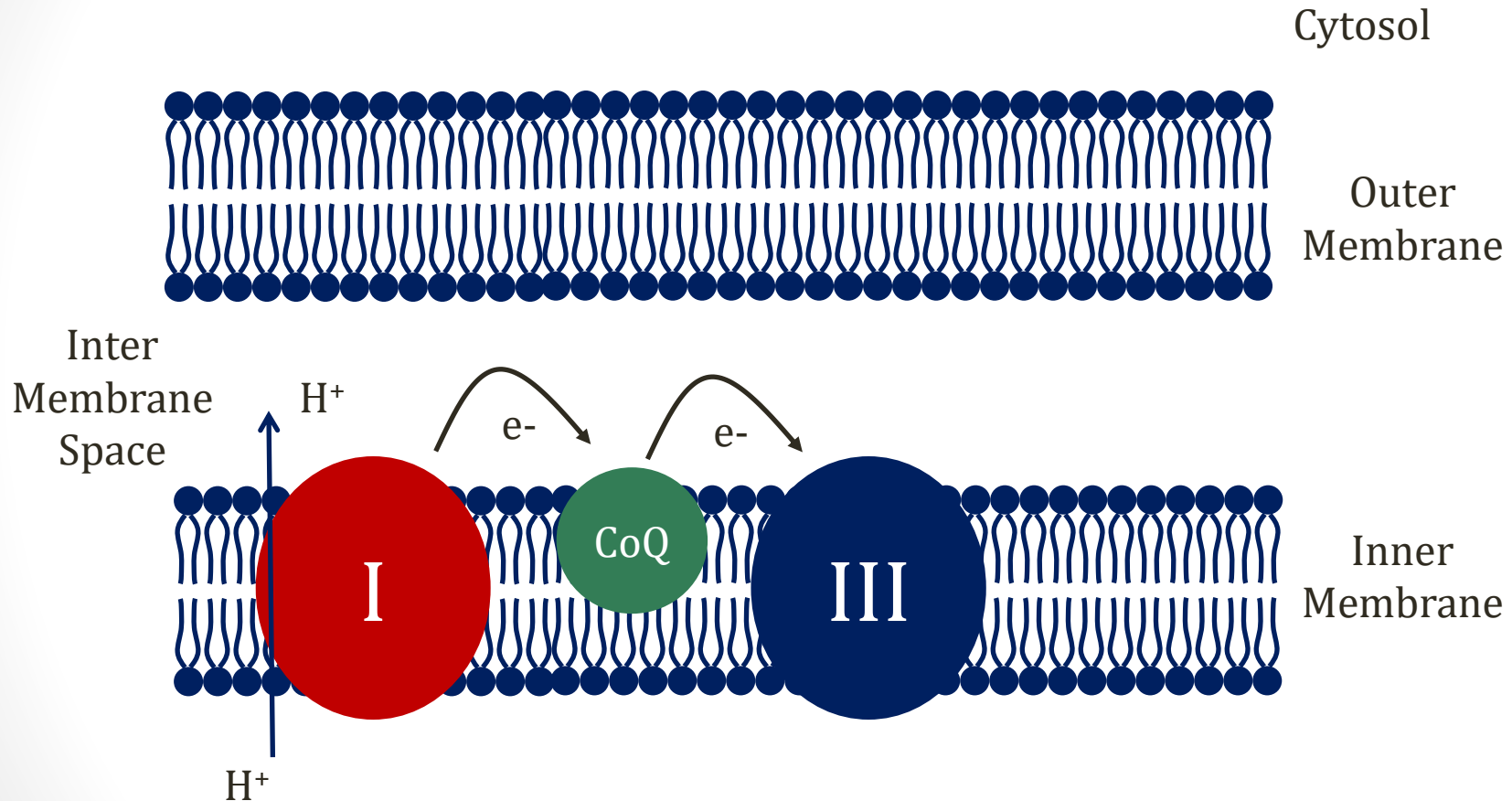


Complex I

- CoQ shuttles electrons to complex III
- Pumps H^+ into intermembrane space
- Key intermediates:
 - Flavin mononucleotide (FMN)
 - Iron sulfur compounds (FeS)



Electron Transport



CoQ 10 Supplements

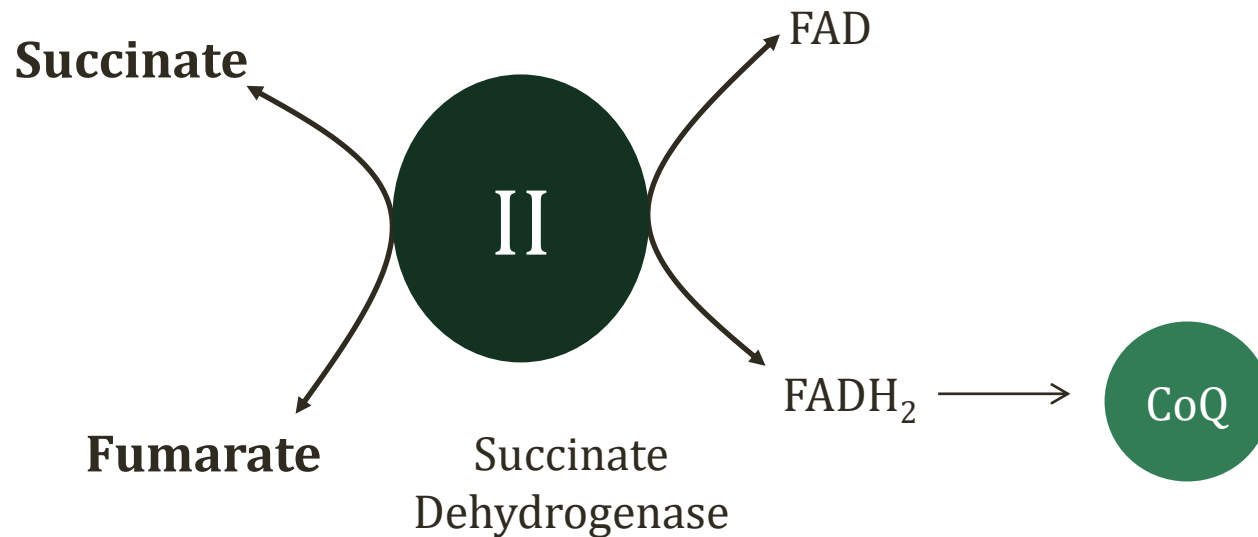
- Some data indicate statins decrease CoQ levels
- Hypothesized to contribute to **statin myopathy**
- CoQ 10 supplements may help in theory
- No good data to support this use



Ragesoss/Wikipedia

Complex II

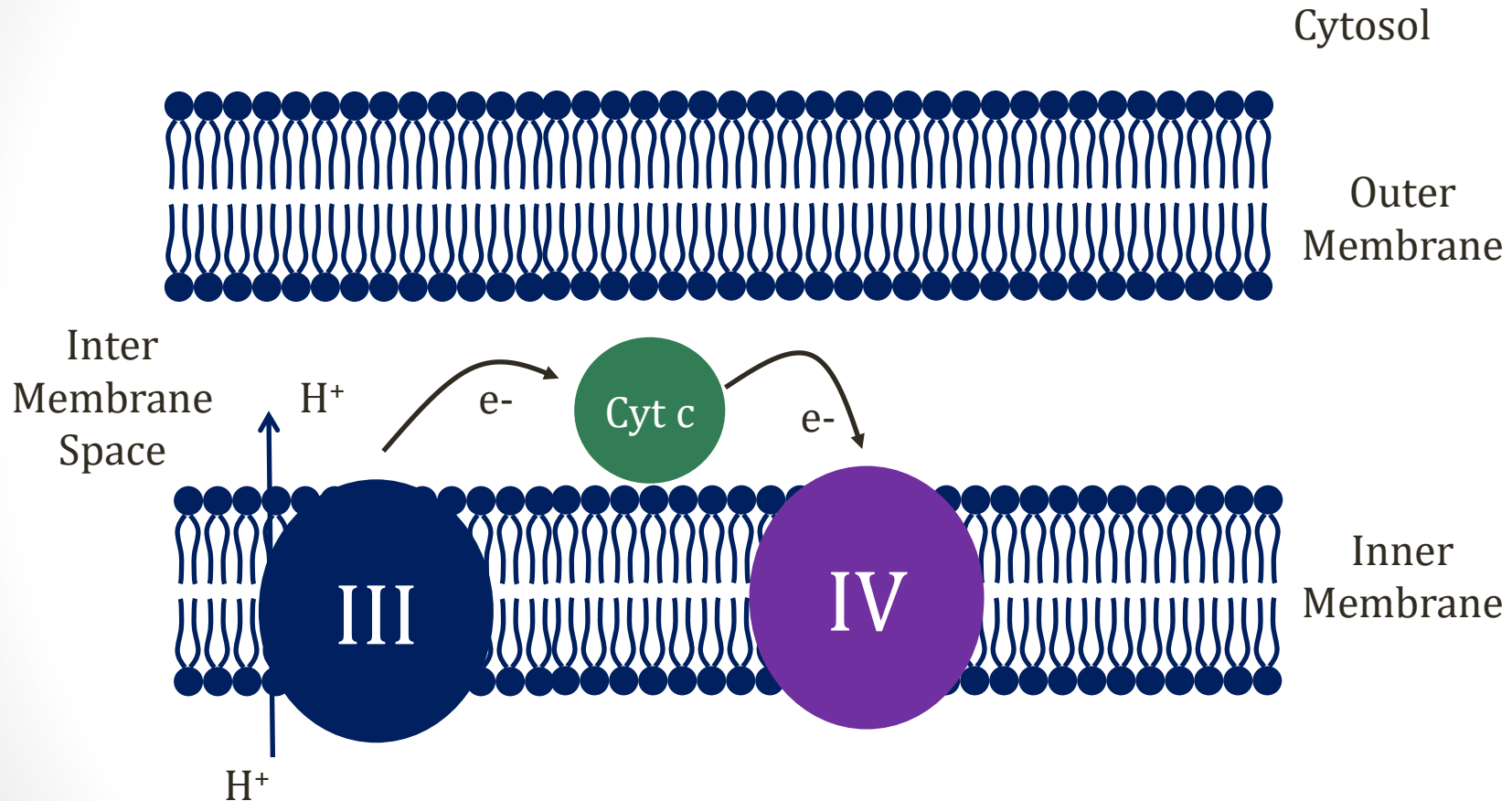
- Succinate dehydrogenase (TCA cycle)
- Electrons from succinate $\rightarrow$ FADH₂ $\rightarrow$ CoQ



Complex III

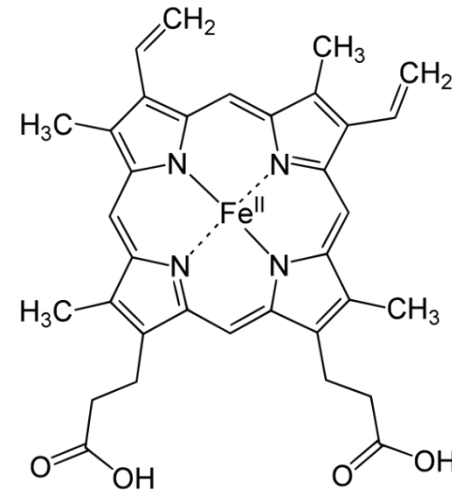
- **Cytochrome bc₁** complex
- Transfers electrons CoQ → **cytochrome c**
- Pumps H⁺ to intermembrane space

Electron Transport



Cytochromes

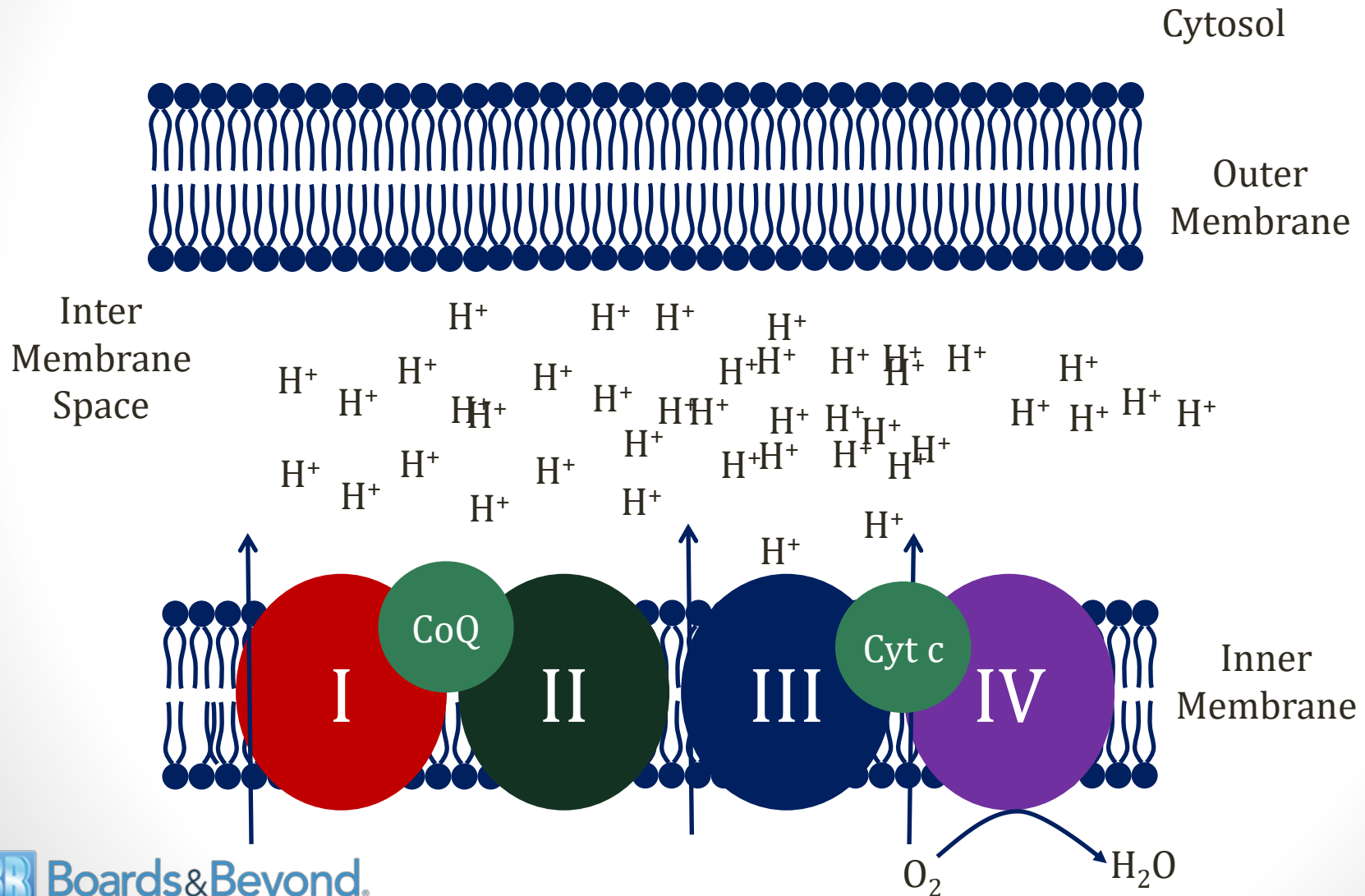
- Class of **proteins**
- Contains a **heme** group
- Iron plus **porphyrin** ring
- Hgb: **mostly** Fe^{2+}
- Cytochromes: $\text{Fe}^{2+} \leftrightarrow \text{Fe}^{3+}$
- Oxidation state changes with electron transport
- Electron transport: a, b, c
- Cytochrome P450: drug metabolism



Complex IV

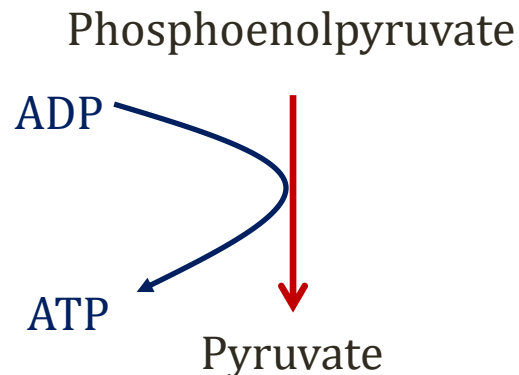
- Cytochrome **a + a₃**
- **Cytochrome c oxidase** (reacts with oxygen)
- Contains **copper** (Cu)
- Electrons and O₂ → H₂O
- Also pumps H⁺

Electron Transport

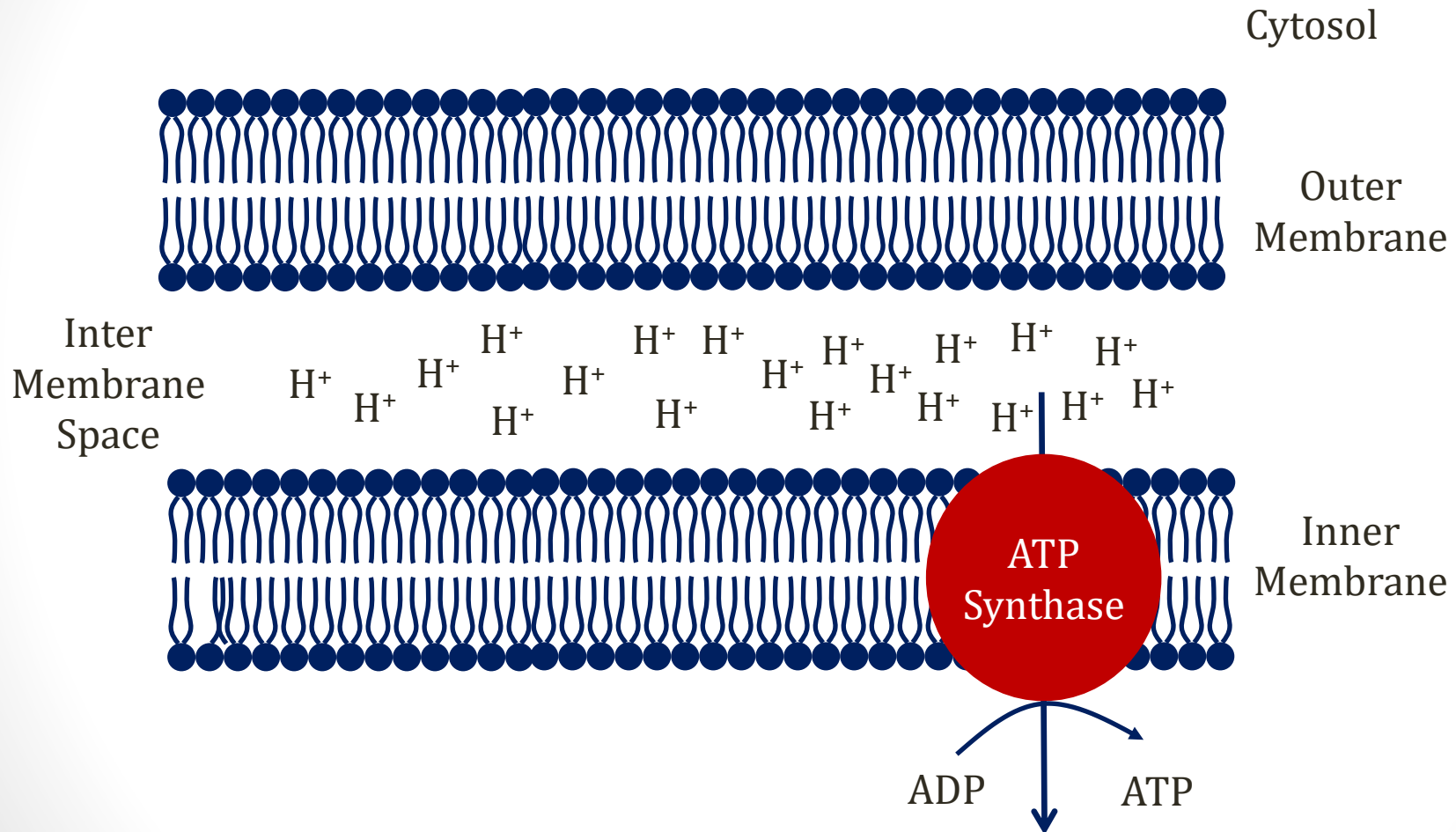


Phosphorylation

- Two ways to produce ATP:
 - Substrate level phosphorylation
 - Oxidative phosphorylation
- Substrate level phosphorylation (via enzyme):



Oxidative Phosphorylation



ATP Synthase

- **Complex V**
- Converts proton (charge) gradient → ATP
 - “electrochemical gradient”
 - “proton motive force”
- Protons move down gradient (“chemiosmosis”)

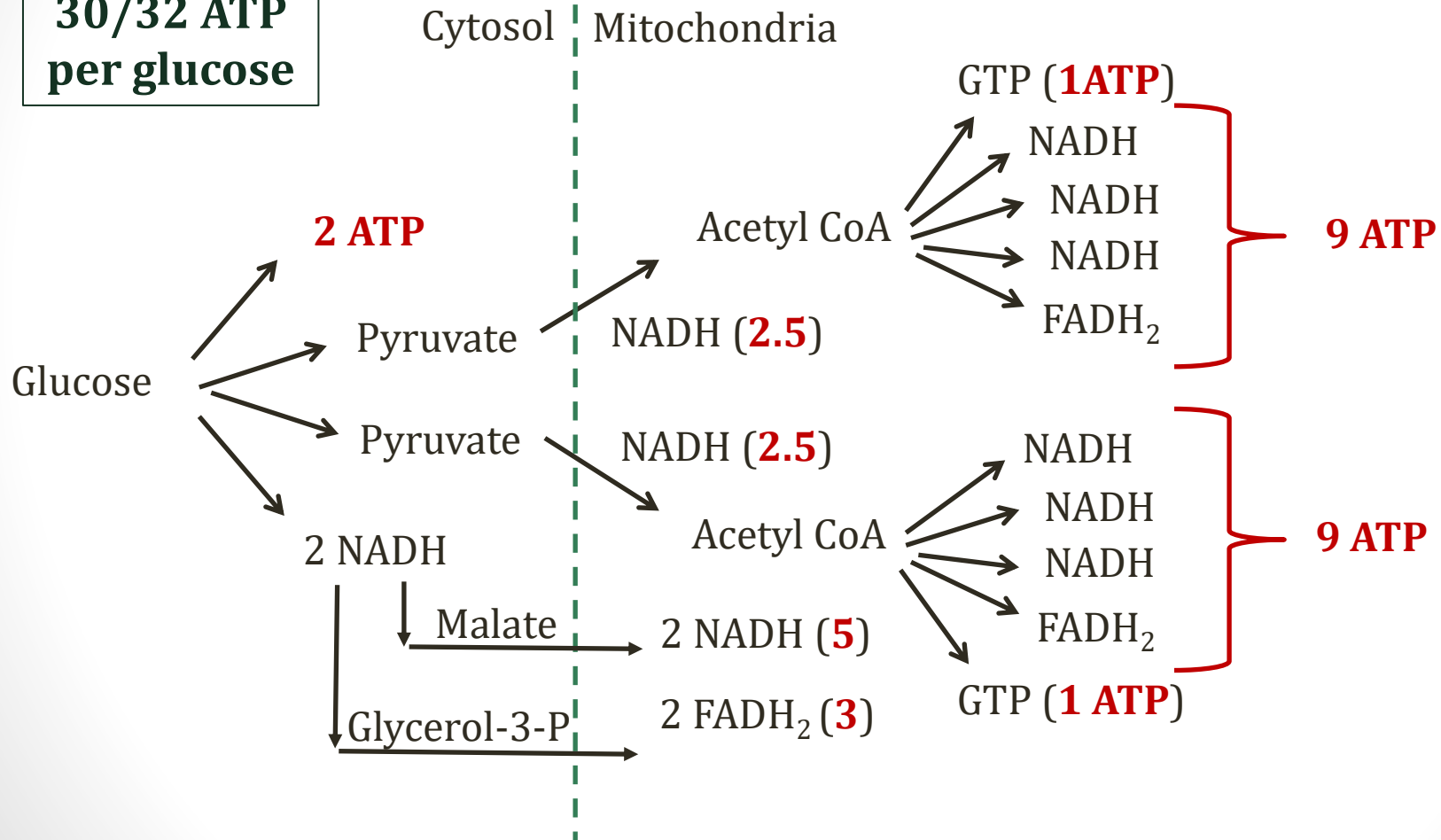
P/O Ratio

- ATP per molecule O_2
- Classically had to be an integer
 - 3 per NADH
 - 2 per $FADH_2$
- Newer estimates
 - 2.5 per NADH
 - 1.5 per $FADH_2$

Hinkle P. P/O ratios of mitochondrial oxidative phosphorylation.
Biochimica et Biophysica Acta 1706 (Jan 2005) 1-11

Aerobic Energy Production

**30/32 ATP
per glucose**



Drugs and Poisons

- Two ways to disrupt oxidative phosphorylation
- #1: Block/inhibit electron transport
- #2: Allow H^+ to leak out of inner membrane space
 - “**Uncoupling**” of electron transport/oxidative phosphorylation

Inhibitors

- Rotenone (insecticide)
 - Binds complex I
 - Prevents electron transfer (reduction) to CoQ
- Antimycin A (antibiotic)
 - Complex III (bc1 complex)
- Complex IV
 - Carbon monoxide (binds a3 in Fe^{2+} state – competes with O_2)
 - Cyanide (binds a3 in Fe^{3+} state)

Cyanide Poisoning

- CNS: Headache, confusion
- Cardiovascular: Initial tachycardia, hypertension
- Respiratory: Initial tachypnea
- **Bright red** venous blood: $\uparrow O_2$ content
- Almond smell
- Anaerobic metabolism: **lactic acidosis**

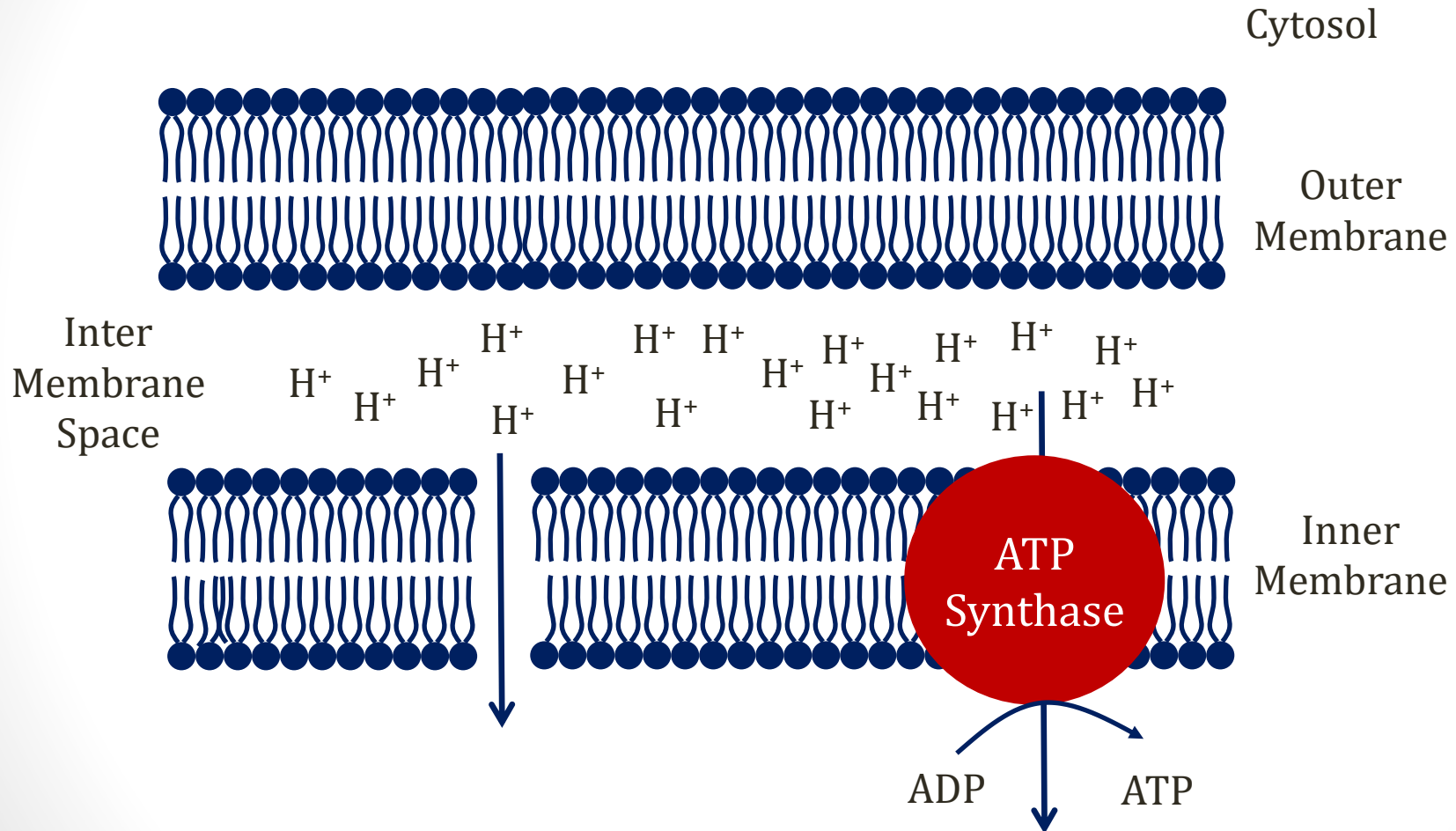


Mullookkaaran/Wikipedia

Cyanide Poisoning

- **Nitroprusside**: treatment of hypertensive emergencies
 - Contains five cyanide groups per molecule
 - Toxic levels with prolonged infusions
- Treatment: **Nitrites** (amyl nitrite)
 - Converts $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ in Hgb (methemoglobin)
 - Fe^{3+} in Hgb binds cyanide, protects mitochondria

Uncoupling Agents



Uncoupling Agents

- **2,4 dinitrophenol (DNP)**
- **Aspirin (overdose)**
- **Brown fat**
 - Newborns (also hibernating animals)
 - **Uncoupling protein 1 (UCP-1, thermogenin)**
 - Sympathetic stimulation (NE, β receptors) $\rightarrow$ lipolysis
 - Electron transport $\rightarrow$ heat (not ATP)

All lead to production of *heat*



Oligomycin A

- Macrolide antibiotic
- **Inhibits ATP synthase**
- Protons cannot move through enzyme
- Protons trapped in intermembrane space
- Oxidative phosphorylation stops
- ATP cannot be generated